

An Introduction to Kivy



Kivy is an open source Python framework, which allows the development of multi-touch mobile and desktop applications with Natural User Interface. Let's take a look at its basics and go through a tutorial on building a basic Kivy application.

Kivy is an open source Python framework for creating cross-platform multi-touch mobile applications with Natural User Interface. Its speed is comparable to other native mobile alternatives like Java for Android and Objective C for iOS. Kivy is the only solution for coding in Python on mobile devices. It is also capable of running on multiple platforms like HTML5 because it does not depend on heavy browser support, and is implemented in C using Cython due to which it runs directly on the GPU. The main aim that was kept in mind while developing Kivy was to run the same code on multiple platforms.

It contains all the modules for building applications such as:

- Modules for input from the mouse, keyboard, TUIO and other OS-specific inputs
- A graphics library
- Multi-touch support provided by a range of widgets
- A designing language, called Kivy Language, to design custom widgets

There are two main dependencies for Kivy to run

on your system:

- Python
- Cython

Kivy installation

To install Kivy on Linux, type the following commands:

```
sudo add-apt-repository ppa:kivy-team/kivy
```

```
sudo apt-get update
```

```
sudo apt-get install python-kivy
```

These commands will install the latest stable version of Kivy on your system.

To install on Windows, take the following steps:

- Download Kivy from <http://www.kivy.org/#download>
 - Once the Kivy Zip file is downloaded, unzip the package in some folder
 - Open that folder and copy the *kivy.bat* file
- Open Windows Explorer and in the address bar, type 'shell:SendTo'.

Paste *kivy.bat* as a shortcut.

This shortcut is used to launch the Kivy application with a right click, as shown in Figure 3.

Kivy Language (KVL)

The Kivy Language is dedicated to creating user interfaces. It separates the presentation logic from the application logic. A Kivy Language file has a *.kv* extension. It features the following.

- **Rules:** These are similar to CSS rules. They can be used for creating interactions and adding presentation logic to the widgets.
- **Root widget:** A Kivy file can have only one root widget. The Kivy Language uses indentations to indicate which rule goes inside the other. The indented line that is the most to the left is the root widget.
- **Dynamic classes:** These were introduced in version 1.7.0. They allow the creation of new widgets and rules without